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Credit Risk Analysis

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Business Problem

Financial institutions such as banks and NBFCs face significant financial losses due to loan defaults. Approving loans without accurately assessing a customer's creditworthiness increases the risk of non-repayment, while overly strict approval criteria may lead to loss of potential customers and revenue.

Objective

The objective of this project is to analyze customer demographic, financial, and credit history data to identify key factors influencing loan default and to build a data-driven approach that helps predict the likelihood of default.

Key Goals

- Identify high-risk customers based on financial and behavioral attributes
- Analyze the impact of factors such as:
  - Income
  - Loan amount
  - Credit history
  - Employment length
- Reduce financial risk for the institution
- Support better loan approval decisions

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df=pd.read_csv("credit_risk_dataset.csv")

df.head()
```

```
   person_age  person_income person_home_ownership  person_emp_length
\
0          22          59000                   RENT              123.0

1          21           9600                    OWN                5.0

2          25           9600               MORTGAGE                1.0

3          23          65500                   RENT                4.0

4          24          54400                   RENT                8.0


  loan_intent loan_grade  loan_amnt  loan_int_rate  loan_status  \
0    PERSONAL          D      35000          16.02            1
1   EDUCATION          B       1000          11.14            0
2     MEDICAL          C       5500          12.87            1
3     MEDICAL          C      35000          15.23            1
4     MEDICAL          C      35000          14.27            1

   loan_percent_income cb_person_default_on_file
cb_person_cred_hist_length
0                  0.59                        Y
3
1                  0.10                        N
2
2                  0.57                        N
3
3                  0.53                        N
2
4                  0.55                        Y
4
```

```python
print(f"Rows: {df.shape[0]}, Columns: {df.shape[1]}")

print("\n--- Data types ---")
print(df.dtypes)

print("\n--- Missing values ---")
print(df.isnull().sum())
```

```python
print("\n--- Target distribution ---")
print(df['loan_status'].value_counts())
print(df['loan_status'].value_counts(normalize=True).round(3))

df['person_emp_length'] =
df['person_emp_length'].fillna(df['person_emp_length'].median())
# Median imputation by loan grade
df['loan_int_rate'] = df.groupby('loan_grade')
['loan_int_rate'].transform(
    lambda x: x.fillna(x.median())
)
print("\n")
print(df.isnull().sum())

# Convert categorical columns properly
df['person_home_ownership'] =
df['person_home_ownership'].astype('category')
df['loan_intent'] = df['loan_intent'].astype('category')
df['loan_grade'] = df['loan_grade'].astype('category')

df.duplicated().sum()

df = df.drop_duplicates()
```

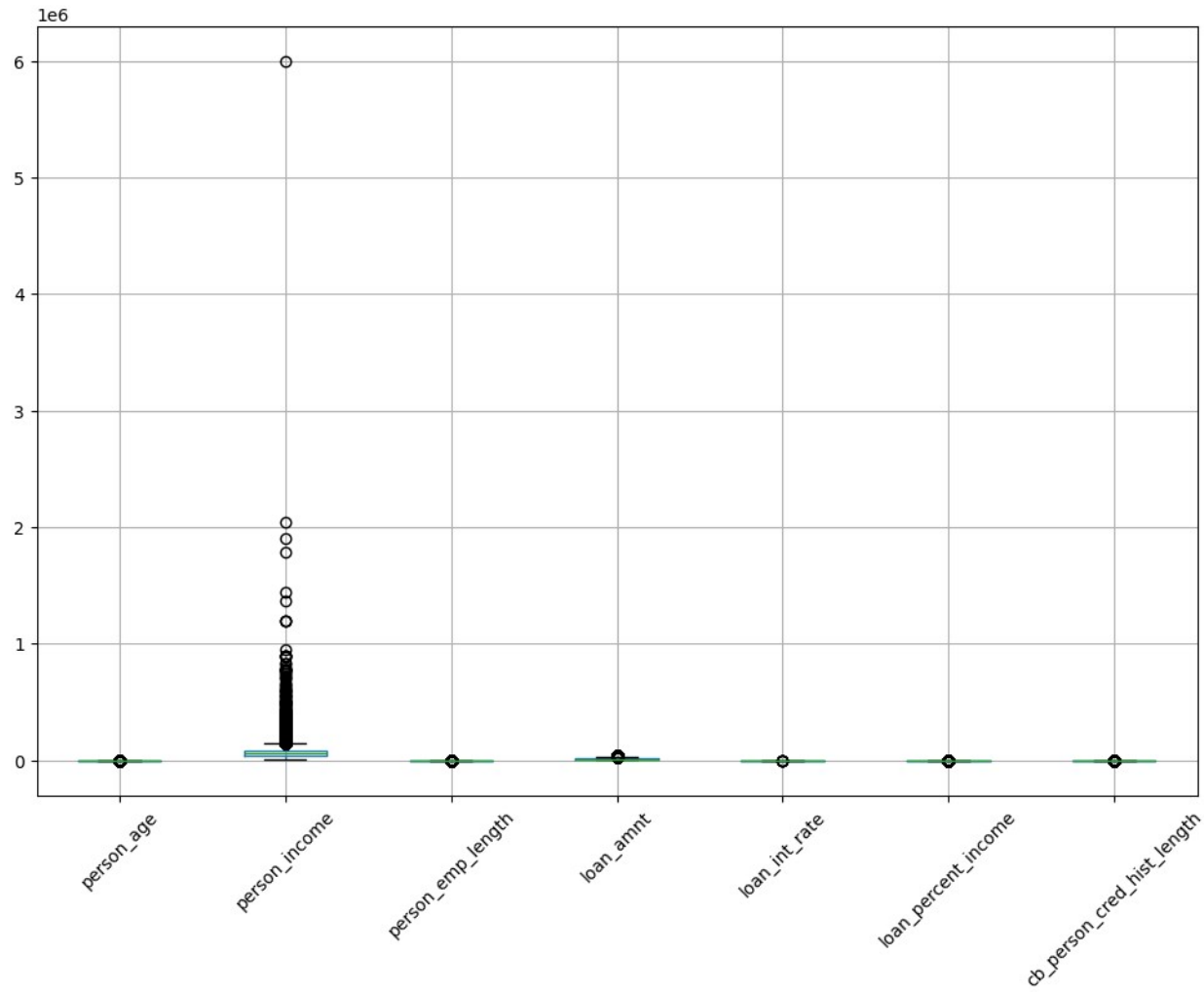
```
Rows: 32581, Columns: 12

--- Data types ---
person_age                      int64
person_income                   int64
person_home_ownership          object
person_emp_length             float64
loan_intent                    object
loan_grade                     object
loan_amnt                       int64
loan_int_rate                 float64
loan_status                     int64
loan_percent_income           float64
cb_person_default_on_file      object
cb_person_cred_hist_length      int64
dtype: object

--- Missing values ---
person_age                        0
person_income                     0
person_home_ownership             0
person_emp_length               895
loan_intent                       0
loan_grade                        0
loan_amnt                         0
loan_int_rate                  3116
```

```
loan_status                       0
loan_percent_income               0
cb_person_default_on_file         0
cb_person_cred_hist_length        0
dtype: int64

--- Target distribution ---
loan_status
0    25473
1     7108
Name: count, dtype: int64
loan_status
0    0.782
1    0.218
Name: proportion, dtype: float64


person_age                     0
person_income                  0
person_home_ownership          0
person_emp_length              0
loan_intent                    0
loan_grade                     0
loan_amnt                      0
loan_int_rate                  0
loan_status                    0
loan_percent_income            0
cb_person_default_on_file      0
cb_person_cred_hist_length     0
dtype: int64
```

```python
#outlier detection

numeric_cols = ['person_age','person_income','person_emp_length',
                'loan_amnt','loan_int_rate','loan_percent_income',
                'cb_person_cred_hist_length']

outliers_dict = {}

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
    outliers_dict[col] = len(outliers)

print(outliers_dict)
```

```
plt.figure(figsize=(12,8))
df[numeric_cols].boxplot()
plt.xticks(rotation=45)
plt.show()


# Age filter
df = df[df['person_age'] <= 100]

df.loc[df['person_age'] > 100, 'person_age'] = 100

# Employment length
df = df[(df['person_emp_length'] >= 0) & (df['person_emp_length'] <=
50)]

# Credit history logical rule
df = df[df['cb_person_cred_hist_length'] <= df['person_age']]

# Function for capping
def cap_outliers(col):
    q1 = df[col].quantile(0.25)
    q3 = df[col].quantile(0.75)
    iqr = q3 - q1
    upper = q3 + 1.5 * iqr
    return np.where(df[col] > upper, upper, df[col])

# Apply capping
cols_to_cap = ['person_income', 'loan_amnt', 'loan_percent_income']

for col in cols_to_cap:
    df[col] = cap_outliers(col)

df.info()
df.describe()
```

```
{'person_age': 1491, 'person_income': 1478, 'person_emp_length': 852,
'loan_amnt': 1679, 'loan_int_rate': 6, 'loan_percent_income': 650,
'cb_person_cred_hist_length': 1139}
```



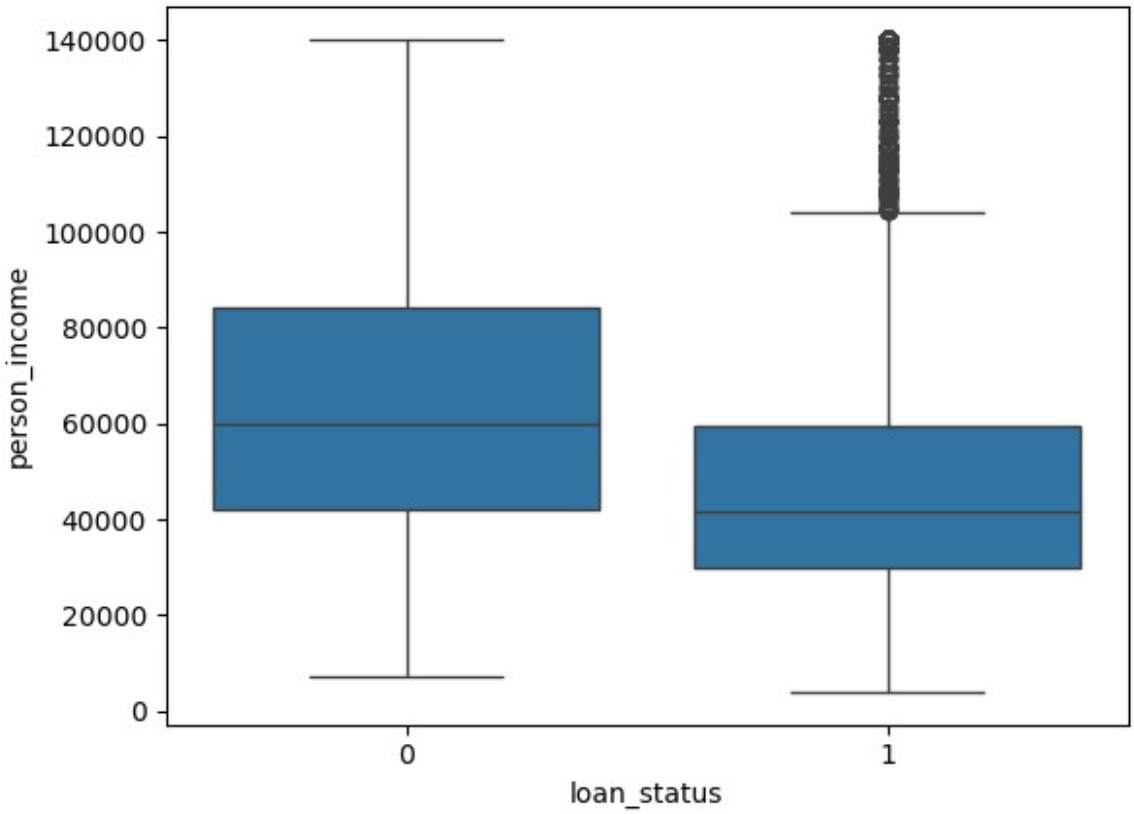

```
<class 'pandas.core.frame.DataFrame'>
Index: 32409 entries, 1 to 32580
Data columns (total 12 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   person_age                  32409 non-null  int64
 1   person_income               32409 non-null  float64
 2   person_home_ownership       32409 non-null  category
 3   person_emp_length           32409 non-null  float64
 4   loan_intent                 32409 non-null  category
 5   loan_grade                  32409 non-null  category
 6   loan_amnt                   32409 non-null  float64
 7   loan_int_rate               32409 non-null  float64
 8   loan_status                 32409 non-null  int64
 9   loan_percent_income         32409 non-null  float64
 10  cb_person_default_on_file   32409 non-null  object
 11  cb_person_cred_hist_length  32409 non-null  int64
```

```
dtypes: category(3), float64(5), int64(3), object(1)
memory usage: 2.6+ MB

         person_age  person_income  person_emp_length     loan_amnt  \
count  32409.000000   32409.000000       32409.000000  32409.000000
mean      27.730754   62427.330649           4.761424   9417.402573
std        6.210445   31800.814798           3.983757   5827.558597
min       20.000000    4000.000000           0.000000    500.000000
25%       23.000000   38500.000000           2.000000   5000.000000
50%       26.000000   55000.000000           4.000000   8000.000000
75%       30.000000   79200.000000           7.000000  12250.000000
max       94.000000  140250.000000          41.000000  23125.000000

       loan_int_rate   loan_status  loan_percent_income  \
count   32409.000000  32409.000000         32409.000000
mean       11.020010      0.218705             0.168857
std         3.213459      0.413374             0.102382
min         5.420000      0.000000             0.000000
25%         7.880000      0.000000             0.090000
50%        10.990000      0.000000             0.150000
75%        13.480000      0.000000             0.230000
max        23.220000      1.000000             0.440000

       cb_person_cred_hist_length
count                   32409.000000
mean                        5.811194
std                         4.057899
min                         2.000000
25%                         3.000000
50%                         4.000000
75%                         8.000000
max                        30.000000
```

Exploratory data analysis

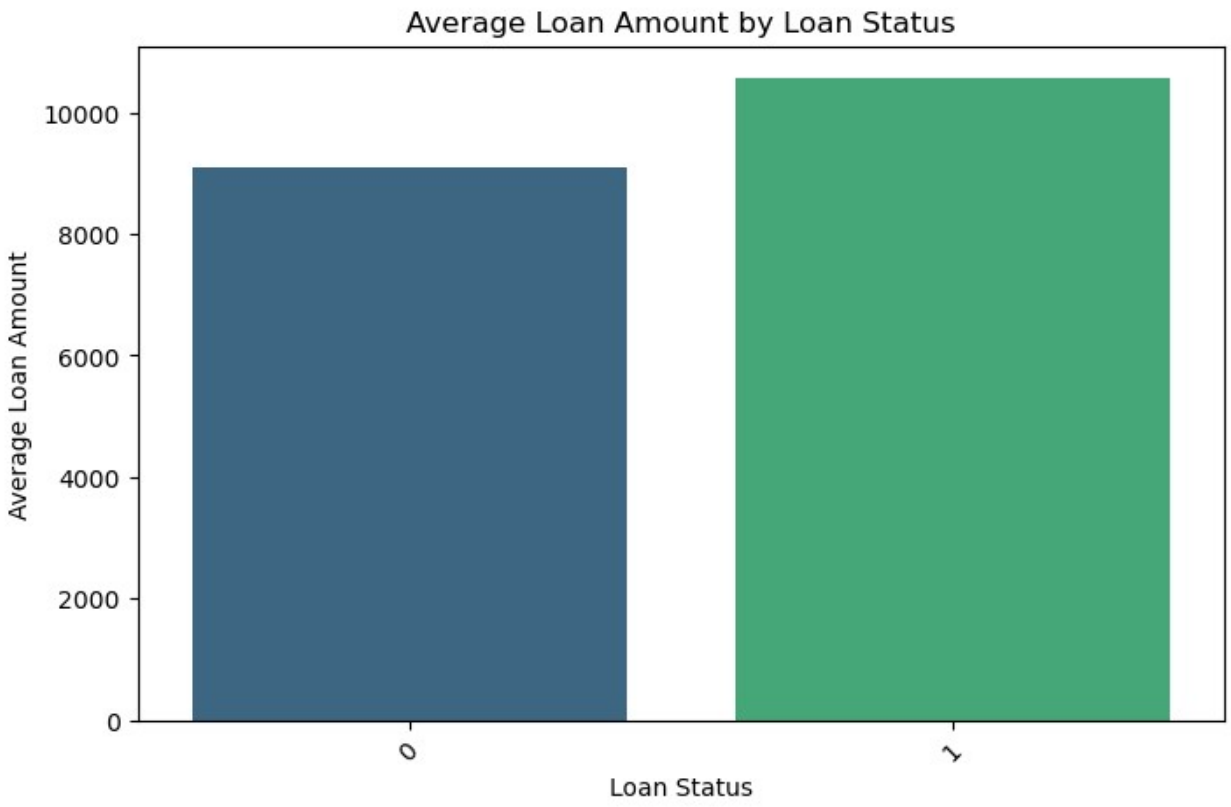
```
#Default Rate Analysis
default_rate = df['loan_status'].mean()
print("Default Rate:", default_rate)

#Segment-wise Default Rate
df.groupby('person_home_ownership')
['loan_status'].mean().sort_values(ascending=False)

#Income vs Default
df.groupby('loan_status')['person_income'].mean()

sns.boxplot(x='loan_status', y='person_income', data=df)
plt.show()

Default Rate: 0.2187046807985436
```

```
C:\Users\hp\AppData\Local\Temp\ipykernel_2820\2515933610.py:6:
FutureWarning: The default of observed=False is deprecated and will be
changed to True in a future version of pandas. Pass observed=False to
retain current behavior or observed=True to adopt the future default
and silence this warning.
  df.groupby('person_home_ownership')
['loan_status'].mean().sort_values(ascending=False)
```

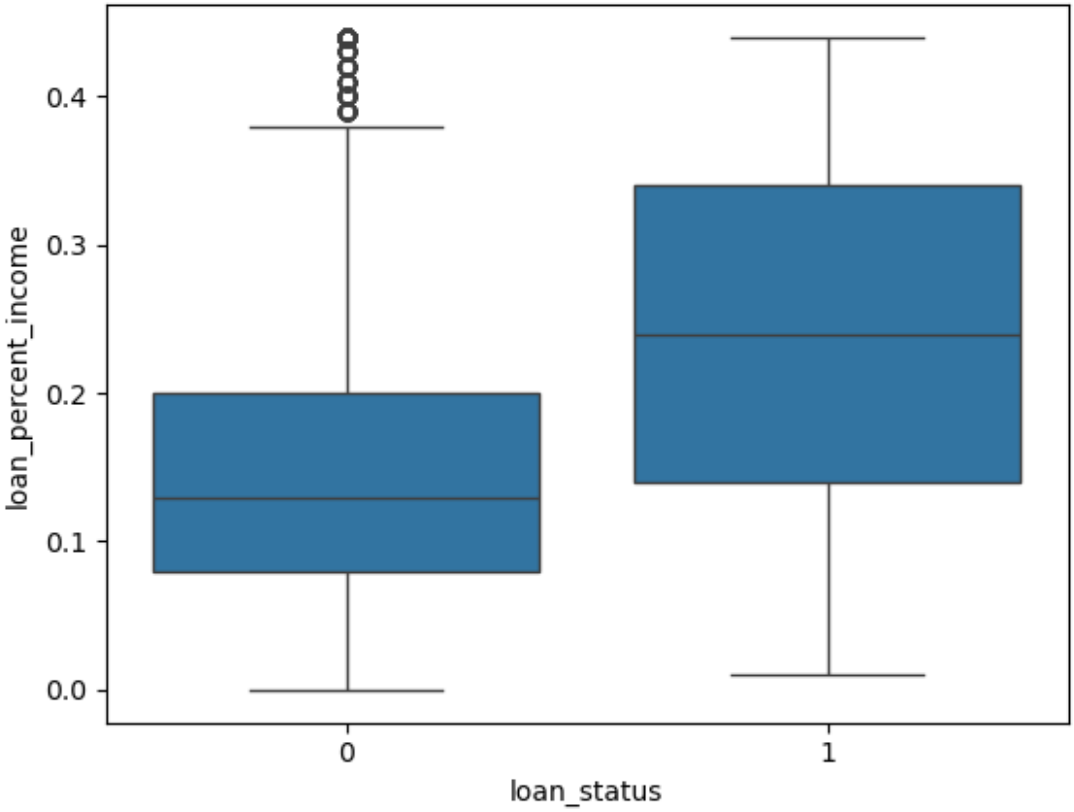



```
loan_means=df.groupby('loan_status')['loan_amnt'].mean()

plt.figure(figsize=(8,5))
sns.barplot(x=loan_means.index, y=loan_means.values,
palette="viridis")

plt.title("Average Loan Amount by Loan Status")
plt.xlabel("Loan Status")
plt.ylabel("Average Loan Amount")
plt.xticks(rotation=45)
plt.show()

C:\Users\hp\AppData\Local\Temp\ipykernel_2820\3875255231.py:4:
FutureWarning:
```

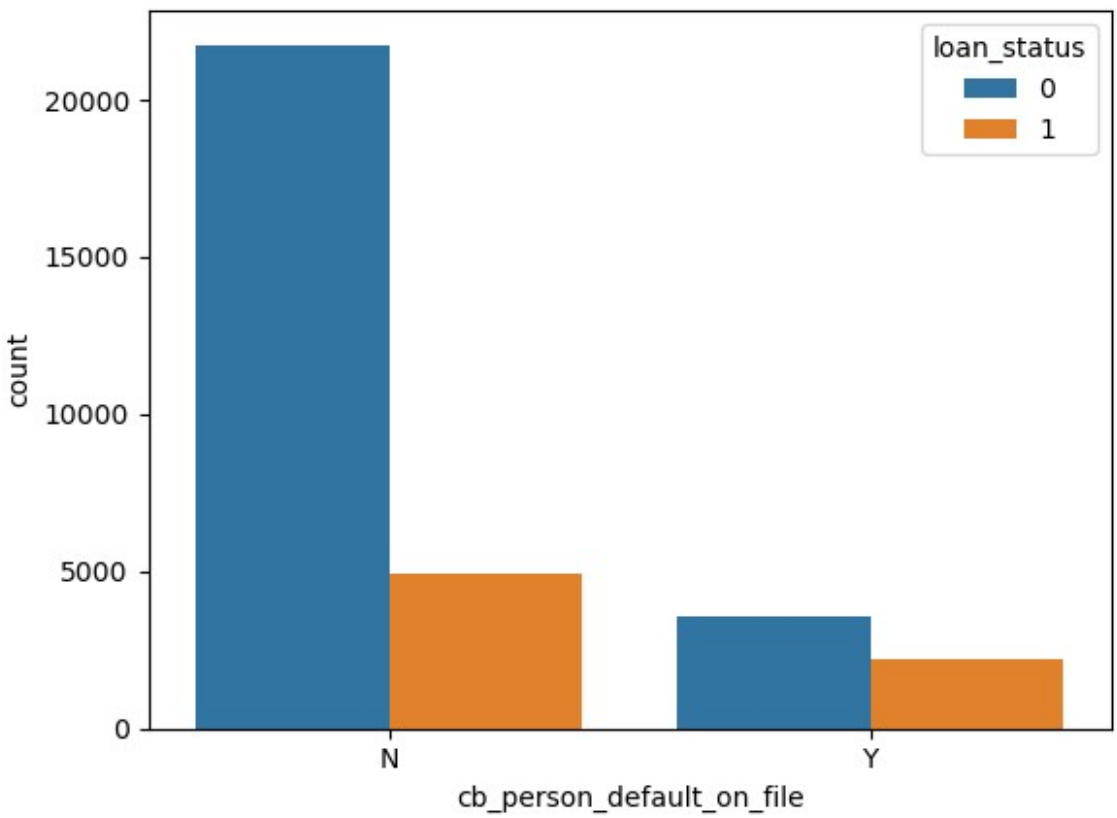
```
Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x=loan_means.index, y=loan_means.values,
palette="viridis")
```




```
#Loan-to-Income Ratio
df.groupby('loan_status')['loan_percent_income'].mean()

sns.boxplot(x='loan_status', y='loan_percent_income', data=df)
plt.show()
```



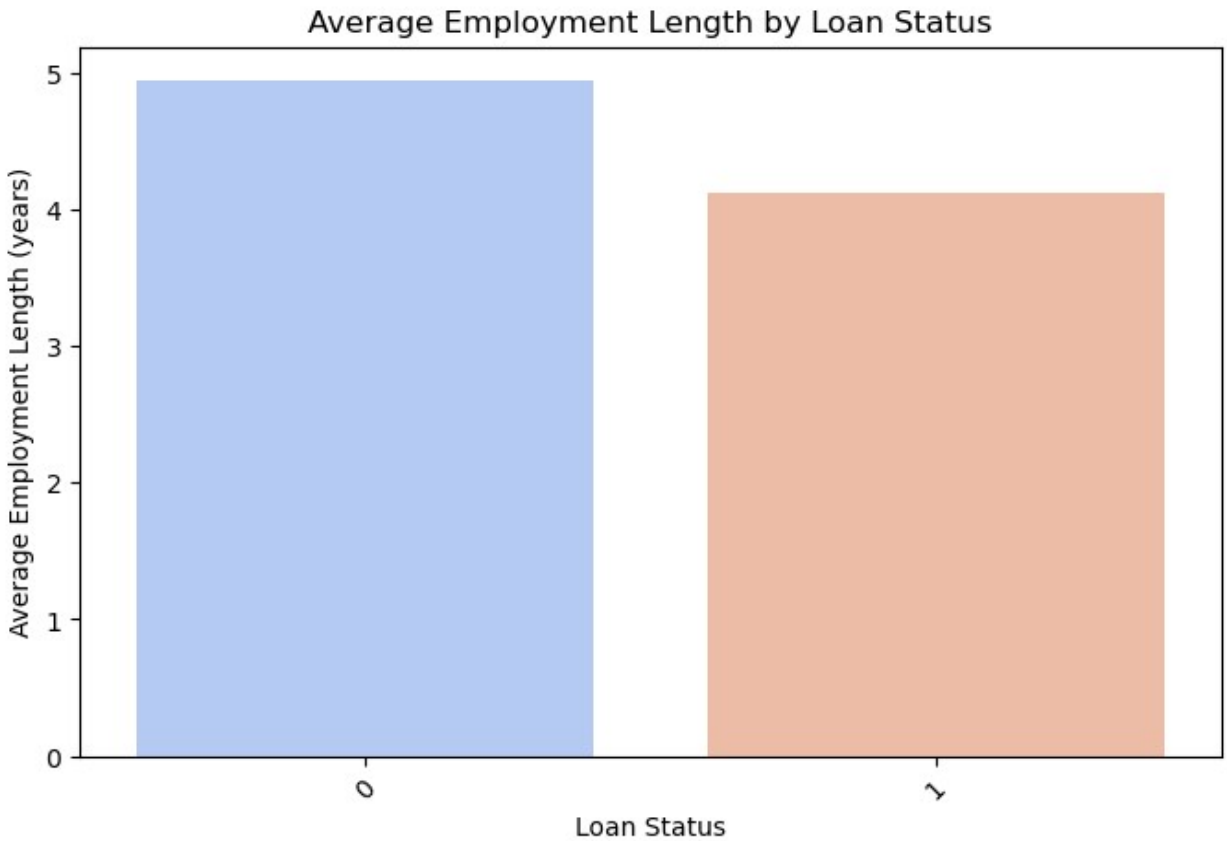

```
#Previous Default Impact

df.groupby('cb_person_default_on_file')['loan_status'].mean()

sns.countplot(x='cb_person_default_on_file', hue='loan_status',
data=df)
plt.show()
```




```
#Employment Length Analysis

emp_means = df.groupby('loan_status')['person_emp_length'].mean()

plt.figure(figsize=(8,5))
sns.barplot(x=emp_means.index, y=emp_means.values, palette="coolwarm")

plt.title("Average Employment Length by Loan Status")
plt.xlabel("Loan Status")
plt.ylabel("Average Employment Length (years)")
plt.xticks(rotation=45)
plt.show()
#Loan Intent Risk Analysis

intent_risk = df.groupby('loan_intent')
['loan_status'].mean().sort_values(ascending=False)

plt.figure(figsize=(8,5))
sns.barplot(x=intent_risk.index, y=intent_risk.values,
palette="magma")

plt.title("Default Rate by Loan Intent")
plt.xlabel("Loan Intent")
plt.ylabel("Default Rate")
```

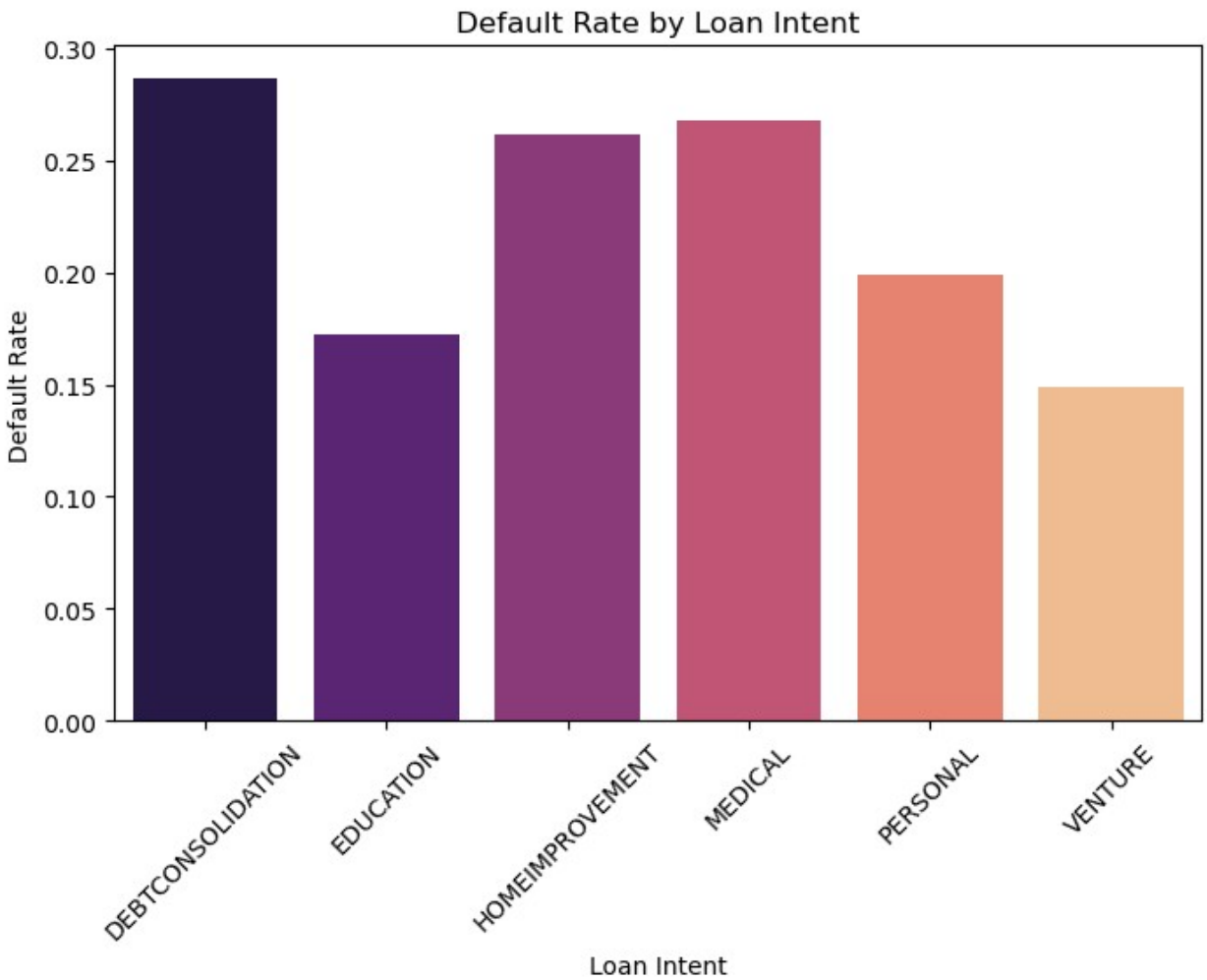
```
plt.xticks(rotation=45)
plt.show()
```

```
C:\Users\hp\AppData\Local\Temp\ipykernel_2820\187976491.py:6:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x=emp_means.index, y=emp_means.values,
palette="coolwarm")
```




```
C:\Users\hp\AppData\Local\Temp\ipykernel_2820\187976491.py:15:
FutureWarning: The default of observed=False is deprecated and will be
changed to True in a future version of pandas. Pass observed=False to
retain current behavior or observed=True to adopt the future default
and silence this warning.
  intent_risk = df.groupby('loan_intent')
['loan_status'].mean().sort_values(ascending=False)
C:\Users\hp\AppData\Local\Temp\ipykernel_2820\187976491.py:18:
FutureWarning:
```

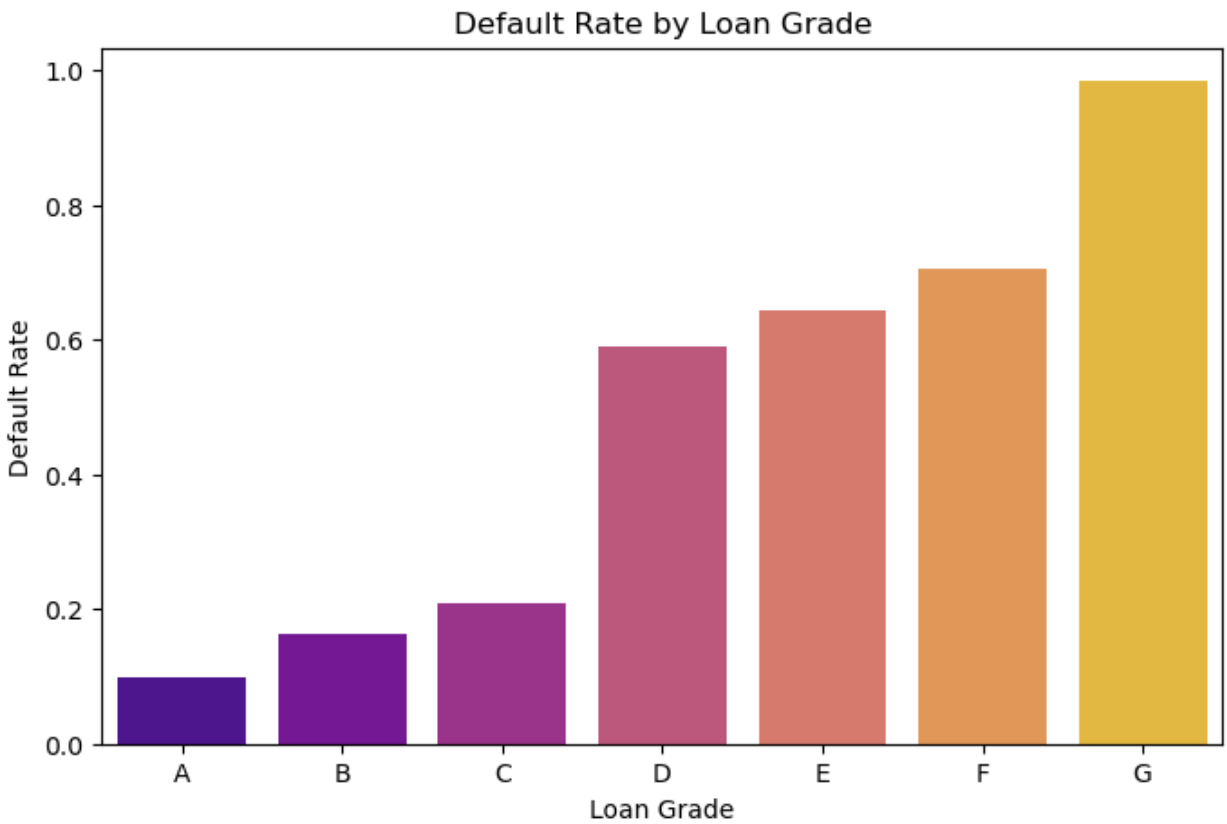
```
Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x=intent_risk.index, y=intent_risk.values,
palette="magma")
```




```
# Calculate default rate by loan grade
grade_risk = df.groupby('loan_grade')
['loan_status'].mean().sort_values(ascending=False)

# Plot
plt.figure(figsize=(8,5))
sns.barplot(x=grade_risk.index, y=grade_risk.values, palette="plasma")

plt.title("Default Rate by Loan Grade")
plt.xlabel("Loan Grade")
```

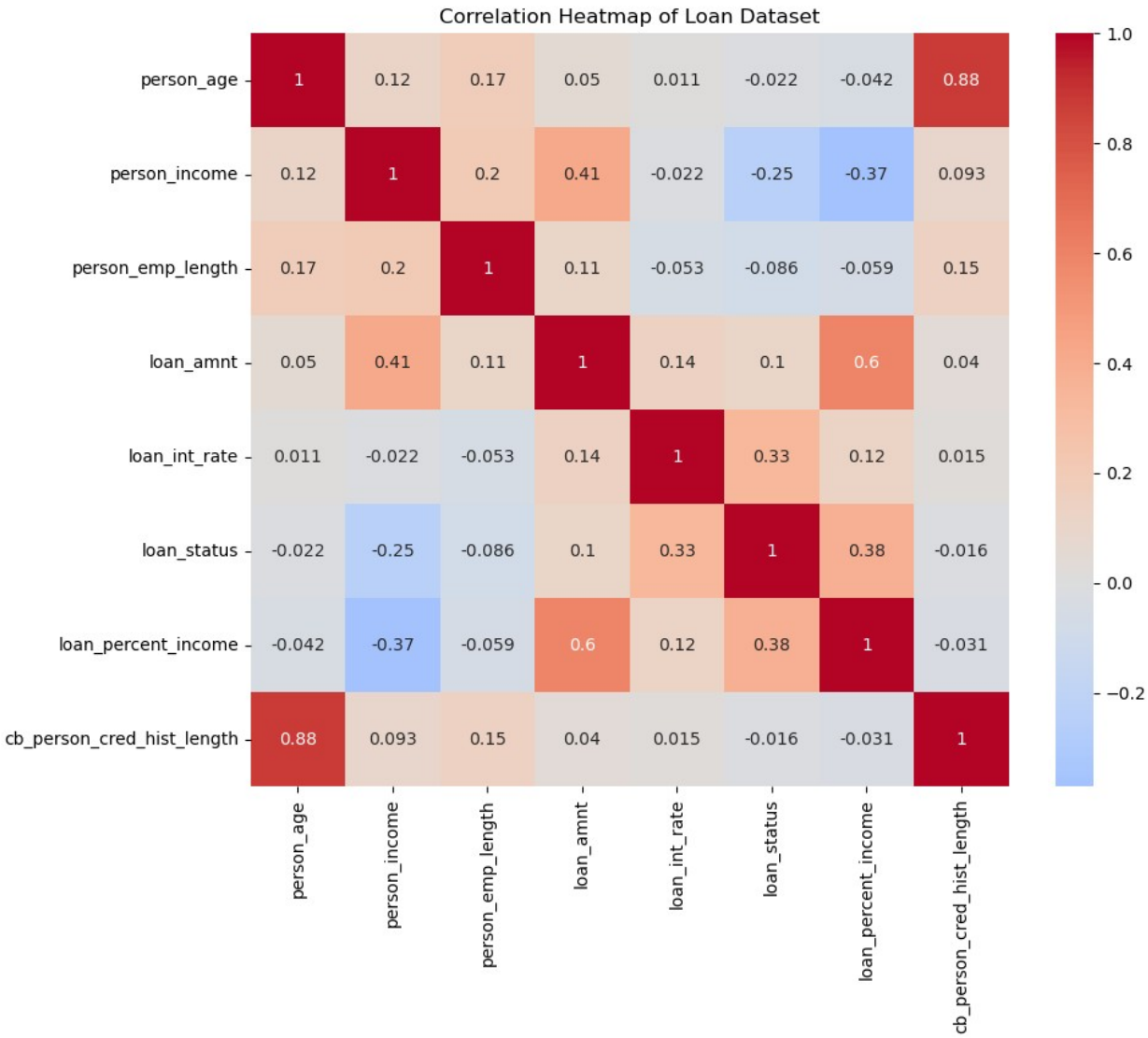
```
plt.ylabel("Default Rate")
plt.show()
```

```
C:\Users\hp\AppData\Local\Temp\ipykernel_2820\572869298.py:2:
FutureWarning: The default of observed=False is deprecated and will be
changed to True in a future version of pandas. Pass observed=False to
retain current behavior or observed=True to adopt the future default
and silence this warning.
  grade_risk = df.groupby('loan_grade')
['loan_status'].mean().sort_values(ascending=False)
C:\Users\hp\AppData\Local\Temp\ipykernel_2820\572869298.py:6:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x=grade_risk.index, y=grade_risk.values,
palette="plasma")
```




```
# Correlation with loan_status
corrs = df.corr(numeric_only=True)
['loan_status'].sort_values(ascending=False)
```

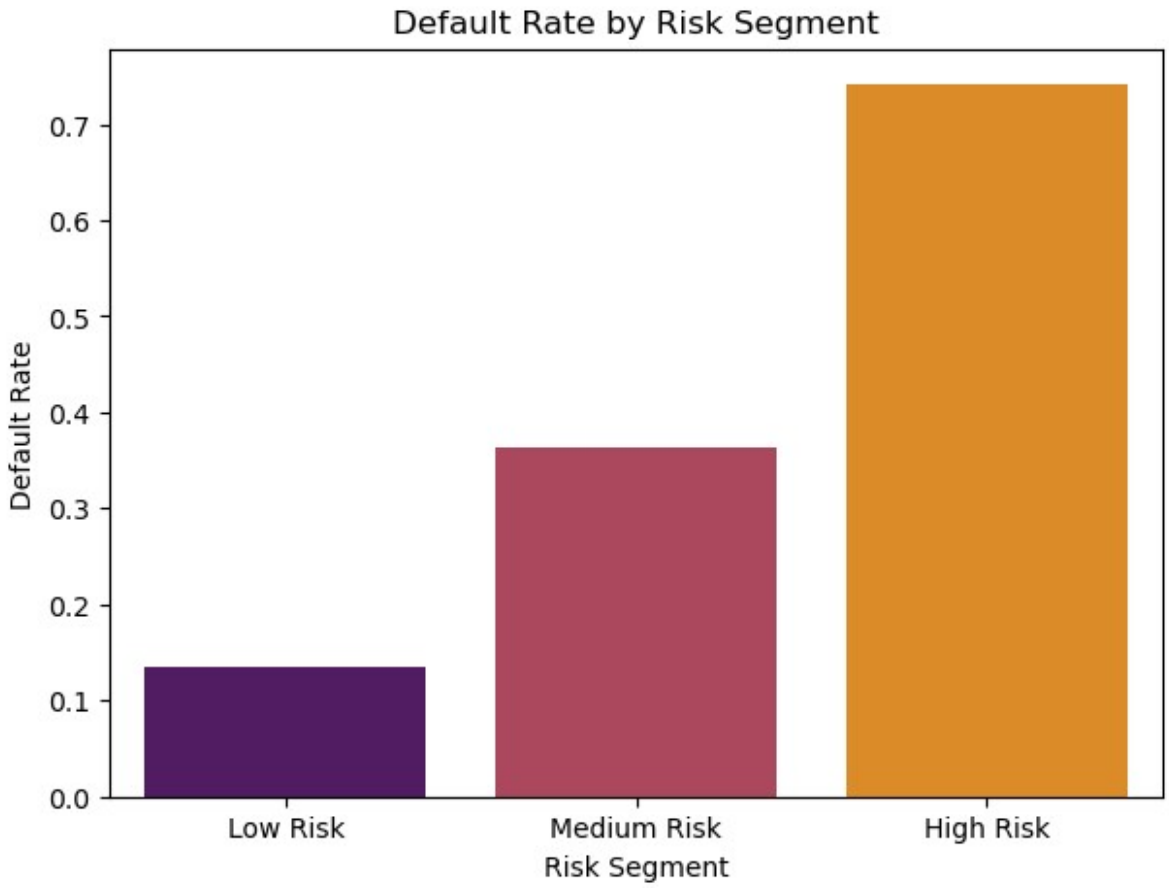
```
# Plot
corr_matrix = df.corr(numeric_only=True)

# Plot heatmap
plt.figure(figsize=(10,8))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", center=0)

plt.title("Correlation Heatmap of Loan Dataset")
plt.show()
```




```
df['risk_segment'] = pd.cut(df['loan_percent_income'],
                            bins=[0, 0.2, 0.4, 1],
                            labels=['Low Risk', 'Medium Risk', 'High
Risk'])
```

```python
# Group by risk segment and calculate default rate
risk_means = df.groupby('risk_segment')['loan_status'].mean()

# Plot
plt.figure(figsize=(7,5))
sns.barplot(x=risk_means.index, y=risk_means.values,
palette="inferno")

plt.title("Default Rate by Risk Segment")
plt.xlabel("Risk Segment")
plt.ylabel("Default Rate")
plt.show()
```

```
C:\Users\hp\AppData\Local\Temp\ipykernel_2820\3808843736.py:6:
FutureWarning: The default of observed=False is deprecated and will be
changed to True in a future version of pandas. Pass observed=False to
retain current behavior or observed=True to adopt the future default
and silence this warning.
  risk_means = df.groupby('risk_segment')['loan_status'].mean()
C:\Users\hp\AppData\Local\Temp\ipykernel_2820\3808843736.py:10:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x=risk_means.index, y=risk_means.values,
palette="inferno")
```



Key Insights

Customers with high loan-to-income ratio have significantly higher default rates Customers with previous defaults are high-risk applicants Low-income groups are more likely to default Certain loan purposes (e.g., personal loans) carry higher risk Lower loan grades are strongly associated with defaults

 Business Recommendations

Reject high loan-to-income ratio customers Apply stricter rules for customers with past defaults Offer smaller loans to low-income applicants Adjust interest rates based on risk segment